StuDy

PREMIUM NOTES

RPSC — General Science Notes

RPSC Test Series — StuDy

High-yield facts, formulas and trap-radar — made for the real exam hall.

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RPSC — General Science Notes

Physics — Core Concepts

- Laws of motion: 1st (inertia), 2nd (F = ma, momentum), 3rd (action–reaction). Momentum p = mv; impulse = change in momentum.
- Work = force × displacement (joule); Power = work/time (watt, 1 HP = 746 W); Energy — KE = ½mv², PE = mgh. 1 unit electricity = 1 kWh = 3.6×10^6 J.
- Light: speed in vacuum 3×10^8 m/s; refraction (bending — pencil in water), total internal reflection (optical fibres, mirage, diamond sparkle), scattering (blue sky, red sunset). Mirror formula 1/v + 1/u = 1/f; lens power in dioptres.
- Sound: speed in air ~332 m/s (increases with temperature); SONAR uses ultrasonics (>20 kHz); audible range 20 Hz–20 kHz; echo needs ≥17 m distance.
- Heat: mercury thermometer (freezing −39°C); absolute zero = −273.15°C = 0 K; latent heat of fusion of ice 80 cal/g. Black surfaces = best absorbers/radiators.

Chemistry — Everyday Science

- Acids & bases: litmus (acid → red, base → blue); turmeric turns red with base; phenolphthalein pink in base. Antacids contain $Mg(OH)_2$/$Al(OH)_3$. Tooth decay starts below pH 5.5.
- Common compounds: washing soda $Na_2CO_3 \cdot 10H_2O$, baking soda $NaHCO_3$, caustic soda NaOH, bleaching powder $CaOCl_2$, plaster of Paris $CaSO_4 \cdot \frac{1}{2}H_2O$, gypsum $CaSO_4 \cdot 2H_2O$, lime CaO, slaked lime $Ca(OH)_2$.
- Metals: most abundant metal in earth's crust — Aluminium; in human body (metal) — Calcium. Stainless steel = Fe + Cr + Ni. Amalgam = alloy with mercury. German silver has NO silver (Cu+Zn+Ni).
- Carbon forms: diamond (hardest natural), graphite (conductor, lubricant), fullerene (C-60 buckyball), graphene. Dry ice = solid CO_2. Laughing gas = N_2O. Tear gas = chloropicrin based.
- Fuels: LPG = propane + butane; CNG = methane mainly; biogas ≈ methane 55–65% + CO_2. Hydrogen = highest calorific value fuel.

Biology — Human Body

- Cells: powerhouse = mitochondria; protein factory = ribosome; brain of cell = nucleus; photosynthesis site = chloroplast (only plants). DNA double helix — Watson & Crick; units = nucleotides.
- Blood: RBC (no nucleus in mammals except camel/llama exception asked sometimes), lifespan 120 days, formed in red bone marrow, destroyed in spleen ('graveyard of RBC'); WBC = soldiers; platelets = clotting. Blood groups — O negative = universal donor, AB positive = universal recipient.
- Organs: largest organ = skin; largest gland = liver (also largest internal organ); strongest muscle (relative) = jaw/masseter; smallest bone = stapes (ear); total bones 206 (adult), 300+ at birth.
- Systems: insulin from pancreas β-cells (diabetes mellitus = insulin deficiency/resistance); dialysis = artificial kidney function; ECG = heart, EEG = brain; normal BP 120/80 mmHg; normal body temp 37°C (98.6°F).
- Plant science: photosynthesis equation $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$; xylem = water transport, phloem = food transport; stomata = transpiration & gas exchange.

Environment & Everyday Tech

- Greenhouse gases: CO_2, CH_4, N_2O, water vapour, CFCs, O_3. Ozone layer in stratosphere; Montreal Protocol (1987) for ODS; Kyoto (1997), Paris Agreement (2015, limit well below 2°C).
- Biodegradable vs non: paper/leaves vs plastics; eutrophication = nutrient enrichment of lakes (algal bloom → oxygen depletion). Biomagnification — DDT classic example.
- Communicable diseases & vectors: malaria (female Anopheles), dengue & chikungunya (Aedes), filaria (Culex), plague (rat flea). Vaccines: BCG (TB), OPV/IPV (polio), DPT, Hepatitis-B.
- Everyday tech: remote controls use infrared; microwave ovens heat via water-molecule vibration (~2.45 GHz); refrigerators — refrigerant cycle; LED = light-emitting diode (less energy than CFL, no mercury).

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